

ALEX MORGAN

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PROFESSIONAL SUMMARY

Results-driven **Software & Systems Engineer** with strong foundations in full-stack web development, API architecture, and intelligent automated systems. Experienced in designing scalable backend microservices using **Python**, **C/C++**, and **JavaScript**, as well as developing high-performance hardware-software IoT pipelines. Proven track record of optimizing system performance by up to 35% through clean code practices, async processing, and reliable test-driven development.

TECHNICAL SKILLS

Languages:	Python, C, C++, JavaScript (ES6+), SQL, Verilog, HTML5/CSS3
Frameworks & Tools:	FastAPI, Flask, Node.js, Express, React.js, PyTorch, NumPy, Pandas
Databases & Cloud:	PostgreSQL, MongoDB, Redis, Docker, AWS (EC2, S3), Git/GitHub, Linux (Bash)
Domains & Architecture:	RESTful APIs, Microservices, IoT Protocols (MQTT, HTTP), System Design, Digital Logic

FEATURED PROJECTS

AI-Powered Intelligent Chatbot System

Python, FastAPI, NLP, PostgreSQL, Docker

- Architected** an end-to-end conversational AI backend using FastAPI and NLP modules, accurately processing 1,000+ daily intent classification queries with **94% accuracy**.
- Engineered** an asynchronous response generation pipeline that reduced end-to-end API response latency by **40%** through optimized Redis caching.
- Containerized** application services using Docker and orchestrated deployment via AWS EC2 for seamless horizontal scalability.

IoT Smart Automation & Control Hub

Python, C++, MQTT, Node.js, Hardware Sensors

- Developed** a real-time environmental monitoring system using embedded microcontrollers, delivering continuous sensor data streams via lightweight MQTT protocols.
- Built** an interactive React and Node.js web dashboard supporting full remote device management with sub-**200ms round-trip control latency**.
- Implemented** automated anomaly detection logic, triggering immediate alerts and reducing unhandled device downtime by **25%**.

Automated Hardware Toll Booth Logic System

Digital Logic, Verilog, C++, Embedded Systems

- Designed and simulated** a complete finite state machine using sequential logic (D Flip-Flops) and logic gate ICs to model complex traffic flow and barrier controls.
- Integrated** IR proximity sensors and servo motor controllers with embedded microprocessors, achieving **99.8% accurate vehicle detection** across simulated field trials.

EXPERIENCE

Software Engineering Intern

June 2025 – August 2025

TechSolutions Inc. | San Francisco, CA

- Collaborated** within an agile team of 5 software engineers to refactor core legacy Python services into high-concurrency microservices, boosting pipeline throughput by **30%**.
- Authored** robust automated unit and integration test suites achieving **85%+ code coverage**, significantly reducing regression defects across bi-weekly deployment cycles.
- Designed and documented** 12+ RESTful API endpoints following OpenAPI/Swagger standards to establish clean integration contracts between front-end UI components and database infrastructure.

EDUCATION

Bachelor of Technology in Computer Science & Engineering

Expected Graduation: May 2026

State University

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Computer Networks, Operating Systems, Digital Logic Design, Object-Oriented Programming, Artificial Intelligence.

CERTIFICATIONS & HONORS

- **AWS Certified Cloud Practitioner** – Amazon Web Services (2025)
- **Meta Back-End Developer Specialization** – Coursera Certification (2025)
- **1st Place Winner** – University Annual Hackathon (Built a real-time intelligent tracking application)